

Project Report

Project Title #25

Android App Development using GenAI Skill-Exchange (Self-Employment)

Project Number	25
Platform	Android (Kotlin + XML Layouts)
Domain	Rural Economy / Self-Employment / Social Commerce
Category	Android Application
Report Type	Internship / Academic Project Report

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1. Introduction

Overview

Skill Exchange is a native Android application designed to connect people who want to learn, teach, and exchange practical skills. The app enables users to create profiles, post learning needs, browse requests, make offers, and track collaboration history.

Purpose

The project supports peer learning and self-employment by giving users a structured platform to discover useful skills and collaborate without relying only on cash-based transactions.

2. Problem Statement And Vision

Problem Statement

Many skilled workers and learners need help from others but do not have a simple platform to discover local skills, request support, or build trust through completed exchanges.

Vision

Skill Exchange aims to create a simple, accessible, and community-focused Android app where users can exchange skills, build reputation, and participate in practical peer learning.

3. Features

Core Features

- User registration and login
- Profile creation and update
- Post skill-learning requests
- Browse recent needs
- Offer help on posted requests
- View collaboration history
- Settings and profile management

UI Update

The current version uses a refreshed teal, coral, and warm neutral color scheme with redesigned splash, login, dashboard, cards, chips, toolbar styling, and feed items.

4. Technology Stack

Android Stack

- Kotlin
- Android SDK
- XML layouts
- Material Components / Material 3
- Room database
- MVVM architecture
- Gradle Kotlin DSL

Development Environment

The app is built and run using Android Studio. It can be tested on an Android emulator or a physical Android device.

5. System Flow

User Flow

A user launches the app, signs in or registers, reaches the dashboard, posts a learning need, browses open requests, offers help, manages profile details, and views history records.

Architecture

The project follows a modular Android structure with data, repository, UI fragments, adapters, view models, and utility classes.

6. Testing And Emulator Notes

Testing

The debug APK builds successfully using Gradle. The project can be installed with `gradlew installDebug` when an emulator or device is connected.

Recommended Emulator

For smoother performance, use Medium Phone, Pixel 4, or Pixel 5 with API 35 or 36, 2048 MB RAM, 8 GB internal storage, and a Google APIs x86_64 image.

7. Conclusion

Conclusion

Skill Exchange demonstrates Android app development using Kotlin, XML layouts, Room, MVVM, and Material styling. The app provides a practical foundation for peer learning and community skill collaboration.

Repository

GitHub: <https://github.com/shvetz45/Skill-Exchange-app>